

1. PROJECT BACKGROUND

This project is developed as a group assignment for the course FCC124 – Introduction to Programming at Albukhary International University (AIU). The objective of the assignment is to demonstrate the ability to develop computer programs using a structured and modular approach in the C programming language (Course Learning Outcome: CLO3).

The chosen project title is the Student Grade Management System (SGMS). This system is a console-based application designed to manage academic records for a small cohort of students. It allows users to add student profiles, input subject grades, compute averages, assign letter grades, and perform various management operations such as editing, deleting, and identifying the top-performing student.

The system is built entirely in C and makes deliberate use of the programming constructs required by the assignment rubric, including arrays, functions with prototypes, call-by-value parameter passing, and both global and local variable scoping.

1.1 Project Scope

- A maximum of 5 students can be stored at any one time.
- 5 fixed subjects: Economics, English, Management, Psychology, and Programming.
- Student IDs are auto-generated starting from 26101001.
- Grades are validated to be within the range 0–100.
- Letter grades follow a standard grading scale (A/B/C/D/F).

1.2 Development Environment

- Language: C

2. PROBLEM STATEMENT

In many educational institutions, student grades are still managed manually or by using general tools such as spreadsheets. Although these methods are commonly used, they can be inefficient and may lead to mistakes, especially when handling a large number of student records. Therefore, there is a need for a simple, reliable, and user-friendly system that can help instructors or administrators manage student grade information more effectively.

The Student Grade Management System is developed to address several common challenges:

1. Entering grades manually can be time consuming and may lead to calculation errors, especially when calculating average marks for multiple subjects.

2. Without an automated system, assigning letter grades based on average scores may become inconsistent.
3. Searching for or updating a specific student's information can be difficult and time-consuming when records are stored manually.
4. Identifying the top-performing student requires checking and comparing all student records one by one.
5. Deleting student records while keeping the data properly organized can also be challenging without a structured management system.

To overcome these issues, the proposed system provides a menu-driven interface with dedicated functions for adding, viewing, editing, deleting, and ranking student records. In addition, the program automatically calculates average marks and assigns letter grades, helping to improve accuracy, efficiency, and overall data management.

2.1 Objectives

- Develop a functional C program that stores and manages student records in memory.
- Automatically compute the average grade across all five subjects for each student.
- Assign letter grades (A, B, C, D, F) based on calculated averages.
- Identify and display the highest-achieving student.
- Demonstrate all four required coding features: arrays, functions, parameters, and scope.